

Nested loops:

➤ Loop inside a loop is called nested loop.

Types of Scenarios:-

1) while :
while :

3) for :
for :
outer loop
inner loop

2) while:
for :

4) for:
while

Patterns:

Pattern 1:

	C1	C2	C3	C4	C5
R1	*				
R2	*	*			
R3	*	*	*		
R4	*	*	*	*	
R5	*	*	*	*	*

Observation

1
2
3
4
5
⋮
n

Step 1: Outer loop will run equals to the total no. of rows.

Step 2: Try to run a inner loop and relate it with rows.

Step 3: print star (*)

no. of stars = row no.

```
for row in range(5):
    for col in range(row+1):
```

```

for col in range(row+1):
    print("*", end=" ")

```

```

for row in range(1,6):
    for col in range(row):
        print("*", end=" ")
    print("\n")

```

DRY RUN

Outer Loop (row)	Inner Loop (col)	Action	Output on screen
row = 1	col = 0	print("*", end=" ")	*
	(inner loop ends)	print("\n")	*\n
row = 2	col = 0	print("*", end=" ")	*
	col = 1	print("*", end=" ")	**
	(inner loop ends)	print("\n")	**\n
row = 3	col = 0	print("*", end=" ")	*
	col = 1	print("*", end=" ")	**
	col = 2	print("*", end=" ")	***
	(inner loop ends)	print("\n")	***\n
row = 4	col = 0	print("*", end=" ")	*
	col = 1	print("*", end=" ")	**
	col = 2	print("*", end=" ")	***
	col = 3	print("*", end=" ")	****
	(inner loop ends)	print("\n")	****\n
row = 5	col = 0	print("*", end=" ")	*
	col = 1	print("*", end=" ")	**
	col = 2	print("*", end=" ")	***
	col = 3	print("*", end=" ")	****
	col = 4	print("*", end=" ")	*****
	(inner loop ends)	print("\n")	*****\n

PATTERN OUTPUT

After row = 1

*

After row = 2

**

After row = 3

After row = 4

After row = 5

Pattern 2:

	C ₁	C ₂	C ₃	C ₄	C ₅
R ₁	*	*	*	*	*
R ₂	*	*	*	*	
R ₃	*	*	*		
R ₄	*	*			
R ₅	*				

Observation

5	→	6-1
4	→	6-2
3	→	6-3
2	→	6-4
1	→	6-5

→ 6 - row number

```

for row in range(1,6):
    for col in range(6-row):
        print("*")
    print()

```

Pattern 3:

	C ₁	C ₂	C ₃	C ₄	C ₅
R ₁	*	*	*	*	*

Observation:-

5

```

for row in range(1,6):

```

	C ₁	C ₂	C ₃	C ₄	C ₅
R ₁	*	*	*	*	*
R ₂	*	*	*	*	*
R ₃	*	*	*	*	*
R ₄	*	*	*	*	*
R ₅	*	*	*	*	*

Observation:-

5
5
5
5
5

for row in range(1,6):

for col in range(1,6):

print("*")

print

Every Row have 5 stars